

End-of-outbreak decision-making under superspreading and onset-date delays

The 2018 Équateur Province (DRC) Ebola outbreak

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Abstract

Deciding when an outbreak is over means estimating the probability that further cases will follow if control measures are relaxed. That estimate depends on two modelling choices that are usually made implicitly: the mechanism through which heterogeneity in transmission is represented, and whether the observed symptom-onset series is read as a series of infections. We compare five renewal models of the 2018 Équateur Province Ebola outbreak, all driven by the same onset-to-onset serial interval, and evaluate under each the risk of additional cases (RAC) after a given day. Overdispersion acting at the level of the day, the transmission event and the individual gives materially different risks at the same nominal dispersion parameter, and the three mechanisms disagree about what that parameter should be when it is estimated. In models that anchor transmission to the infector’s own onset and represent the incubation period explicitly, both decision thresholds are crossed 8–12 days earlier than under the infection-anchored counterparts, and the post-intervention reproduction number is roughly a third of the naive estimate. Separating the risk of additional *cases* from the risk of additional *transmission* quantifies the contribution of individuals already infected but not yet symptomatic, a distinction the infection-anchored models cannot represent.

1 Introduction

An end-of-outbreak declaration is a decision under uncertainty: interventions are withdrawn on the judgement that the probability of further cases has become acceptably small. The standard tool for that judgement is a renewal model fitted to the observed case series and projected forward under a counterfactual in which control is relaxed. Two modelling choices inside that procedure are rarely examined.

The mechanism of overdispersion. Transmission of Ebola virus disease, like that of many pathogens, is highly heterogeneous, and this is usually represented by a single dispersion parameter k . But models can share a mean–variance relationship at the population level while differing in *where* the excess variance sits — in a day’s aggregate incidence, in each transmission event, or in each infected individual. Those are different statements about the world, and we show they imply different end-of-outbreak risks. They also make k a different quantity in each case, so a literature estimate obtained under one mechanism is not portable to another: transplanting the individual-level value $k = 0.18$ into a day-level model is a category error, and one that has been made.

Onset dates are not infection dates. Renewal models are routinely fitted with a symptom-onset time series substituted directly for an infection time series. This conflates two distinct events separated by an incubation period, which for Ebola virus disease has a mean of about 11.4 days. We quantify the consequences by comparing infection-anchored models with explicitly onset-anchored counterparts, in which transmission is referenced to the infector’s own onset and the incubation period is modelled.

We compare five models on a single, well-documented outbreak, all driven by the same onset-to-onset serial interval so that the comparison isolates mechanism and anchoring rather than confounding either with a different assumed interval. Section 2 describes the data and delay distributions, Section 3 the models, Section 4 the risk estimand and its closed forms, and Section 5 the inference. Section 6 reports four analyses and a supplementary comparison. Section 7 lists the sensitivity analyses the design calls for; none has been run, and there is no discussion section at this stage.

2 Data and delay distributions

2.1 The 2018 Équateur outbreak

The observed series is a daily count of Ebola virus disease cases by *symptom-onset date* for the 2018 Équateur Province outbreak in the Democratic Republic of the Congo: 54 cases, with onsets on 31 distinct days between 5 April 2018 and 2 June 2018, and at most 6 onsets on any one day. An Ebola Response Team (ERT) was present from 8 May 2018 and withdrew on 24 July 2018. Indexing days from the first observed onset:

	Day index	Date
First onset (index case)	0	5 April 2018
ERT arrival	33	8 May 2018
Last observed onset	58	2 June 2018
ERT withdrawal (window end)	110	24 July 2018

The analysis window is days 0–110, i.e. 111 days, with 28 cases before the ERT arrived and 26 after. The series is padded with zero counts from the last observed onset to the end of the window. Those zeros are not cosmetic: they are the evidence that drives the risk down, and the estimand is undefined over the decision period without them. No under-reporting adjustment is made.

Day 0 is treated as an initial condition in every model, so likelihoods run over days 1, . . . , 110. Conditioning on a single imported index case asserts that nothing was transmitting or incubating beforehand. For the infection-anchored models this is mild — the renewal recursion simply starts from one case. For the onset-anchored models it is stronger, because it also asserts an *empty incubation pipeline* at day 0: anyone infected shortly before the index case’s onset would contribute onsets in the first weeks that the model can explain only as transmission from the index case, inflating R_{pre} . The onset-anchored models are therefore the ones exposed to a mis-identified index case (Section 7, item 5).

2.2 Serial interval

Two estimates are available for this outbreak. The *generic* Ebola estimate — a discretised gamma with mean 15.3 d and standard deviation 9.3 d — is used throughout, for all five models. The *outbreak-specific* estimate, with mean 19.46 d and standard deviation 6.08 d and fitted to contact-traced pairs from this outbreak, is retained only for a sensitivity analysis.

The reason for the choice is the variance budget of Section 2.3. Splitting an onset-to-onset interval into an incubation period and a time-from-onset-to-transmission (TOST) delay requires the two variances to sum to the serial interval’s. The generic estimate has variance 86.49 d², which comfortably accommodates standard published Ebola incubation estimates. The outbreak-specific estimate has variance 36.97 d², *smaller* than the incubation variance alone (65.61 d²), and is therefore structurally inadmissible for the onset-anchored models. That incompatibility is itself informative about how much structure the decomposition imposes, and it is why the outbreak-specific sensitivity analysis can be run only for the infection-anchored models.

2.3 Discretisation, indexing and the decomposition

Continuous delays are gamma distributed and discretised by the method of Cori et al. (2013): the mass at integer lag s is $\int_{s-1}^{s+1} (1 - |s - u|) g(u) du$, normalised. Three discrete delays are used, and they differ in the lag at which they start:

Distribution	Weights	First lag	Interpretation
Serial interval	w_s	1	onset of infector $\rightarrow$ onset of infectee
Incubation period	f_a^{inc}	1	infection $\rightarrow$ onset
Onset to transmission	f_s^{tost}	0	onset of infector $\rightarrow$ transmission

Write $F_r = \sum_{s=1}^r w_s$ with $F_0 = 0$ for the serial-interval distribution function, and let

$$\bar{f}_r^{\text{tost}} = \Pr(S > r) = 1 - \sum_{s=0}^r f_s^{\text{tost}}, \quad \bar{f}_r^{\text{inc}} = \Pr(A > r) = 1 - \sum_{a=1}^r f_a^{\text{inc}} \quad (1)$$

be the survival functions of the TOST delay S and the incubation period A . Note $\bar{f}_0^{\text{inc}} = 1$ but $\bar{f}_0^{\text{tost}} = 1 - f_0^{\text{tost}} \leq 1$: a case has all of its incubation ahead of it on the day it is infected, but has already had its lag-0 transmission opportunity on the day it becomes symptomatic.

The lag-0 rule is what makes the onset-anchored recursion causal. Because f^{inc} places no mass at lag 0, an infection on day t cannot produce an onset before day $t + 1$, so a day's onset cohort is fully determined before that day's own transmission is evaluated. The TOST distribution *may* place mass at lag 0, so a case can transmit on its own onset day. At most one of the two may be supported at lag 0: if both were, infection, onset and onward transmission could all fall within one day and the recursion would not be well ordered.

The variance budget. For an infector with onset on day u , transmission occurs on day $u + S$ and the infectee's onset on day $u + S + A$, so the onset-to-onset interval is $S + A$ and matching the target serial interval requires

$$\mathbb{E}[S] + \mathbb{E}[A] = 15.3, \quad \text{Var}[S] + \text{Var}[A] = 86.49. \quad (2)$$

The incubation period is fixed from a published Ebola estimate — mean 11.4 d, standard deviation 8.1 d (WHO Ebola Response Team, 2014) — and f^{tost} is the gamma carrying the residual mean and variance, here mean 3.9 d and standard deviation 4.57 d. A negative residual variance would make the incubation estimate inadmissible against the serial interval. Discretising and then convolving is not quite the same operation as convolving and then discretising, so the decomposition is verified numerically: the discrete convolution $f^{\text{tost}} * f^{\text{inc}}$ differs from the discretised target w by 0.00070 in total variation. All delays are truncated at lag 110, the length of the analysis window, so nothing is truncated within it.

Remark. The convolution is faithful, but *how* the mean and variance are split between incubation and TOST is an assumption and is not identified by the serial interval alone. Since it is exactly this split that the onset-anchored models exploit, Section 7, item 2, bounds how much of the onset-anchoring result depends on it.

3 Models

3.1 Notation

Let C_t be the observed count on day t . The three *infection-anchored* (“naive”) models set $I_t := C_t$, reading onset dates as infection dates — the practice under study. The two *onset-anchored* models set $D_t := C_t$ and treat onsets as onsets, with infections and onsets separated by the incubation

period. The reproduction number R_t takes the value R_{pre} before the ERT arrives and R_{post} from its arrival onwards (Section 3.5), and $k > 0$ is a dispersion parameter, small k corresponding to high heterogeneity.

Remark. $\text{NB}(\text{mean} = \mu, \text{disp} = \kappa)$ denotes the negative binomial with mean μ , variance $\mu + \mu^2/\kappa$ and probability generating function $G(s) = (\kappa/(\kappa + \mu(1-s)))^\kappa$; in particular $G(0) = (1 + \mu/\kappa)^{-\kappa}$. $\text{Gamma}(\alpha, \beta)$ has shape α , rate β and mean α/β .

3.2 Infection-anchored models

All four infection-anchored models share one generative skeleton. Cases beget cases through the serial interval: an individual infected on day u produces secondary infections that appear on day $u + \ell$ with probability w_ℓ , and the reproduction number in force on the day a secondary case appears sets how many of them to expect. Write

$$\lambda_t = \sum_{s \geq 1} w_s I_{t-s} \quad (3)$$

for the force of infection driving day t . What distinguishes the models is the offspring law — where, if anywhere, the excess variance sits.

Cori — the Poisson baseline. An individual infected on day u generates, at each lag ℓ and independently of every other, a Poisson number of offspring with mean $R_{u+\ell}w_\ell$. Poisson superposition then gives the day- t total as

$$\text{Cori :} \quad I_t \mid \lambda_t \sim \text{Poisson}(R_t \lambda_t). \quad (4)$$

This is the standard renewal model of Cori et al. (2013) and the model underlying most published end-of-outbreak risk calculations. It has no dispersion parameter, and its natural and inference-convenient forms coincide.

DLO — day-level overdispersion. The negative binomial is applied directly to the day’s aggregate incidence, with the same dispersion on every day:

$$\text{DLO :} \quad I_t \mid \lambda_t \sim \text{NB}(\text{mean} = R_t \lambda_t, \text{disp} = k). \quad (5)$$

This model has no per-individual reading at all. The excess variance is a property of the day rather than of anybody in it, and it does not scale with how many people are infectious — so k here is a statement about day-to-day variability in aggregate incidence, not about how unequally individuals transmit. That is the defining feature, and the reason the same numerical k means something quite different here than in the two models below. Its natural and inference-convenient forms also coincide, and the likelihood is closed form.

SSE — superspreading events. The offspring law is again negative binomial, but with a dispersion that *does* scale with the force of infection:

$$\text{SSE :} \quad I_t \mid \lambda_t \sim \text{NB}(\text{mean} = R_t \lambda_t, \text{disp} = k \lambda_t). \quad (6)$$

That scaling is what makes k an *individual-level* offspring dispersion. Each infector independently contributes $\text{NB}(R_t w_\ell, k w_\ell)$ offspring at lag ℓ ; these all share the success probability $k/(k + R_t)$, which does not depend on ℓ , so by negative binomial closure they pool into exactly (6), and each individual’s own offspring total is $\text{NB}(R, k)$. The decomposition is used again in Section 4.2.

For inference, (6) is un-marginalised into its Poisson–Gamma representation,

$$\tilde{\lambda}_t \mid \lambda_t \sim \text{Gamma}(k \lambda_t, k), \quad I_t \mid \tilde{\lambda}_t \sim \text{Poisson}(R_t \tilde{\lambda}_t), \quad (7)$$

in which the day's transmission opportunity is inflated or deflated by a fresh Gamma factor. For SSE this is optional — the marginal (6) is closed form, making SSE the only one of the five compared models that needs no latent variables — but the same representation is what makes SSE-SO tractable below.

SSI — superspreading individuals. Heterogeneity is attached to people rather than to occasions. Each infected individual carries a latent infectivity drawn independently from $\text{Gamma}(k, k)$, with mean 1 and variance $1/k$, so the aggregate infectivity of the day- t cohort is $\text{Gamma}(kI_t, k)$ and infections arise as a Poisson process driven by past infectivity:

$$\text{SSI:} \quad Y_t \mid I_t \sim \text{Gamma}(kI_t, k), \quad I_t \mid (Y_u)_{u < t} \sim \text{Poisson}(R_t \lambda_t^Y), \quad \lambda_t^Y = \sum_{s \geq 1} w_s Y_{t-s}. \quad (8)$$

Here $I_t = 0$ forces $Y_t = 0$. Marginally each individual's offspring total is $\text{NB}(R, k)$, the same as under SSE — the two models are calibrated to the same individual-level offspring distribution and differ only in mechanism. But the Y_t cannot be integrated out of the joint law, so (8) is both the natural and the inference-convenient form. On this series SSI carries 31 latent infectivities.

Remark (The Poisson limit). As $k \rightarrow \infty$ the negative binomial in (6) converges to the Poisson and the Gamma noise in (8) degenerates to $Y_t = I_t$, so SSE and SSI both collapse onto Cori. DLO loses its excess variance in the same limit, but without the parameter ever having had an individual-level reading to lose.

3.3 Onset-anchored models

Transmission is now referenced to the infector's own symptom onset rather than to their unobserved infection time, which is appropriate for a disease with negligible pre-symptomatic transmission. Two things change. The serial interval is replaced by the TOST distribution, so the force of infection is a weighted sum of past *onset* cohorts; and infection and onset become distinct events, separated by the incubation period. Each of the three models below is the exact onset-anchored counterpart of one of (4), (6) and (8).

Let J_t denote the number of infections on day t and D_t the number of onsets. In all three models the day- u infections are carried forward to their onset days by independent incubation draws,

$$(N_{u,a})_{a \geq 1} \mid J_u \sim \text{Multinomial}(J_u, (f_a^{\text{inc}})), \quad D_t = \sum_{a \geq 1} N_{t-a,a}, \quad (9)$$

and only the onsets are observed: the infection counts J_t and their allocations $N_{u,a}$ are latent. That latent block is large, discrete and unnecessary, because it can be integrated out exactly.

Proposition 1 (Onsets alone suffice). *Suppose that, conditional on a non-negative day- t rate E_t that is a function of $D_{0:t-1}$ and of any latent variables attached to days $u < t$, the infections satisfy $J_t \mid E_t \sim \text{Poisson}(E_t)$, and that they are allocated to onset days by (9). Then that model and*

$$D_t \mid (E_u)_{u < t} \sim \text{Poisson}\left(\sum_{a \geq 1} f_a^{\text{inc}} E_{t-a}\right) \quad (10)$$

define the same joint law for the observed onset process (D_t) .

The proof is in Appendix A. It rests on Poisson thinning, which turns the multinomial split (9) into independent Poisson parts, and on Poisson superposition, which recombines them into (10); the same thinning argument supplies the incubation pipeline of Section 4.3. The hypothesis covers all three models: two have Poisson offspring outright, and SSE-SO's negative binomial is Poisson conditional on its own Gamma factor, exactly as in (7). Each model below is therefore given twice — as it is defined, and as it is fitted.

Cori-SO — the onset-anchored Poisson baseline. A case with onset on day u infects others s days later with probability f_s^{tost} , in $\text{Poisson}(R_{u+s}f_s^{\text{tost}})$ numbers and independently across lags and individuals. Summing over infectors,

$$\text{Cori-SO : } J_t \mid D_{0:t} \sim \text{Poisson}(R_t \lambda_t), \quad \lambda_t = \sum_{s \geq 0} f_s^{\text{tost}} D_{t-s}, \quad (11)$$

together with (9). Proposition 1 applies with $E_t = R_t \lambda_t$ and reduces this to

$$D_t \mid D_{0:t-1} \sim \text{Poisson} \left(\sum_{a \geq 1} f_a^{\text{inc}} R_{t-a} \sum_{s \geq 0} f_s^{\text{tost}} D_{t-a-s} \right), \quad (12)$$

a deterministic recursion in the observed onsets with no latent variables at all.

SSE-SO — superspreading events, onset-anchored. The offspring law is negative binomial with a dispersion scaling with the force of infection, as in (6):

$$\text{SSE-SO : } J_t \mid D_{0:t} \sim \text{NB}(\text{mean} = R_t \lambda_t, \text{disp} = k \lambda_t), \quad \lambda_t = \sum_{s \geq 0} f_s^{\text{tost}} D_{t-s}, \quad (13)$$

together with (9). Writing the negative binomial as a Poisson–Gamma mixture puts a fresh Gamma factor on each day’s transmission opportunity, and Proposition 1 then applies with $E_t = R_t \tilde{\lambda}_t$:

$$\tilde{\lambda}_t \mid \lambda_t \sim \text{Gamma}(k \lambda_t, k), \quad D_t \mid (\tilde{\lambda}_u)_{u < t} \sim \text{Poisson} \left(\sum_{a \geq 1} f_a^{\text{inc}} R_{t-a} \tilde{\lambda}_{t-a} \right). \quad (14)$$

Here, unlike in SSE, the Gamma factor cannot be integrated out again: each D_t pools contributions from many days, each carrying its own independent $\tilde{\lambda}$, so the onsets are *not* marginally negative binomial and there is no closed-form likelihood. SSE-SO carries one latent per day whose TOST-weighted onset sum is non-zero; because the index case contributes $f_t^{\text{tost}} D_0 > 0$ at every lag out to the truncation, that is *every* inference day — 110 latents on this series.

SSI-SO — superspreading individuals, onset-anchored. Each case carries a latent infectivity drawn from $\text{Gamma}(k, k)$, so the day- t onset cohort has aggregate infectivity $\text{Gamma}(k D_t, k)$ and drives a Poisson infection process, as in (8):

$$\begin{aligned} \text{SSI-SO : } Y_t \mid D_t &\sim \text{Gamma}(k D_t, k), & J_t \mid (Y_u)_{u \leq t} &\sim \text{Poisson}(R_t \lambda_t^Y), \\ \lambda_t^Y &= \sum_{s \geq 0} f_s^{\text{tost}} Y_{t-s}, \end{aligned} \quad (15)$$

together with (9). Proposition 1 applies with $E_t = R_t \lambda_t^Y$ and gives

$$Y_t \mid D_t \sim \text{Gamma}(k D_t, k), \quad D_t \mid (Y_u)_{u < t} \sim \text{Poisson} \left(\sum_{a \geq 1} f_a^{\text{inc}} R_{t-a} \sum_{s \geq 0} f_s^{\text{tost}} Y_{t-a-s} \right). \quad (16)$$

The latents are the same cohort infectivities as under SSI — 31 of them here — and, as there, they cannot be integrated out. Section 5.3 describes how the latent blocks of (14) and (16) are handled.

Remark (The Poisson limit, and the role of Cori and Cori-SO). As in Remark 3.2, $k \rightarrow \infty$ collapses SSE-SO and SSI-SO onto Cori-SO. Neither Cori nor Cori-SO is one of the five compared models. They serve as the verification targets for the collapse checks of Section 5.6; Cori is additionally the model whose published Équateur risk curve provides the external check of the risk calculation; and they fix the reference point for Section 4.4, in which a mechanism’s effect on the risk is read as its departure from the Poisson limit.

3.4 The five compared models

Model	Anchoring	Excess variance acts at the level of	Latents
DLO	infections (naive)	the day — NB on aggregate incidence	none
SSE	infections (naive)	the transmission event	none
SSI	infections (naive)	the individual	Y_t
SSE-SO	symptom onsets	the transmission event	$\tilde{\lambda}_t$
SSI-SO	symptom onsets	the individual	Y_t

3.5 When does R switch?

R switches from R_{pre} to R_{post} on day 33 (8 May 2018) **in each model’s own time index**. In the onset-anchored models this is unambiguous: R_t multiplies the TOST-weighted force of infection producing *infections* on day t , so the intervention acts on transmission from that date and the resulting onsets appear an incubation period later. In the infection-anchored models R_t multiplies the force of infection producing *cases appearing* on day t , so the switch attributes the intervention’s effect to onsets from that date.

The displacement between the two conventions is the infectee’s **incubation period**, not the serial interval. A case with onset on day t was infected on day $t - A$ with $A \sim f^{\text{inc}}$; the infection-anchored model dates the intervention’s effect on that transmission to t , the onset-anchored model to $t - A$. The relevant delay is therefore $\mathbb{E}[A] = 11.4$ d. (The serial interval, 15.3 d, is the infector-onset to infectee-onset gap and does not separate the two conventions.)

This asymmetry is a consequence of onset-as-infection conflation rather than a modelling artefact: it is one route by which naive models mis-date an intervention’s effect, and Section 6.3 quantifies what it does to R_{post} and to the risk curve. Section 7, item 4, describes the run that would separate this indexing effect from the structural one.

4 The risk of additional cases and of additional transmission

4.1 Definition

Writing the estimand as $\Pr(\cdot \mid \text{data up to day } t)$ invites a filtering reading that the analysis does not use, so we state it as a procedure.

Fit $(R_{\text{pre}}, R_{\text{post}}, k)$ and all latent states to the complete record, days 0–110. Then, for each day t : retain the inferred state attached to the observed history through day t , discard the realised trajectory after day t , reset R to $R^* = R_{\text{pre}}$ from day $t + 1$ onwards, and simulate a counterfactual future. $\text{RAC}(t)$ is the posterior probability that this replicated future contains at least one further case.

Two things follow that the shorthand hides. The fit is fully retrospective by design: the parameters come from all 111 days, not from the history through t . And for the latent-variable models the *retained state* also comes from the full-data posterior, so it is a *smoothed* state informed by data after day t (Section 4.5).

Under the onset-anchored models a second quantity becomes distinct:

- $\text{RAC}(t)$, the probability of at least one further *case*, i.e. a further entry in the onset series after day t . It counts individuals who are already infected on day t and become symptomatic afterwards, even if no new transmission occurs, so it does not approach zero until the incubation pipeline has drained. This is the directly observable quantity and the headline throughout.
- $\text{RAT}(t)$, the probability of at least one further *transmission event* after day t . It isolates the effect of relaxing interventions but is not directly observable.

Under the three infection-anchored models an infection and the case it produces fall on the same day by construction, so the two coincide *by assumption* — which is precisely the conflation under study. RAT is therefore reported only in the supplementary comparison of Section 6.4.

RAC(t) is one number per day, not a distribution: it is a posterior probability, so the average over draws happens inside it. In each case below we obtain a conditional zero-probability given one posterior draw, and report

$$\text{RAC}(t) = 1 - \frac{1}{M} \sum_{m=1}^M \exp(\log \Pr(\text{no further case} \mid \theta^{(m)})), \quad (17)$$

over M posterior draws, and likewise for RAT. Because this is an average it carries Monte-Carlo error, which is estimated from the between-chain spread and reported beside every curve.

4.2 Infection-anchored models

Define the pooled remaining-transmission weight

$$\Lambda(t) = \sum_{u \leq t} I_u (1 - F_{t-u}) \quad (\text{SSE}, \text{DLO}), \quad \Lambda_Y(t) = \sum_{u \leq t} Y_u (1 - F_{t-u}) \quad (\text{SSI}), \quad (18)$$

where $(1 - F_{t-u})$ is the probability that a case first appearing on day u still has transmission to come after day t ; with $F_0 = 0$, the $u = t$ term enters with full weight.

Proposition 2 (Conditional zero-probabilities, infection-anchored models). *Conditional on one draw of the parameters and, for SSI, the retained latent infectivities,*

$$\Pr(\text{no further case after } t)_{\text{Cori}} = \exp(-R^* \Lambda(t)), \quad (19)$$

$$\Pr(\text{no further case after } t)_{\text{SSE}} = (1 + R^*/k)^{-k \Lambda(t)}, \quad (20)$$

$$\Pr(\text{no further case after } t)_{\text{SSI}} = \exp(-R^* \Lambda_Y(t)), \quad (21)$$

$$\Pr(\text{no further case after } t)_{\text{DLO}} = \prod_{j>t} (1 + R^* \mu_j/k)^{-k}, \quad \mu_j = \sum_{s \geq 1} w_s I_{j-s}, \quad (22)$$

where in (22) the future counts are set to zero, $I_u := 0$ for $u > t$.

Proof. For Cori, each observed case contributes $\text{Poisson}(R^* w_\ell)$ offspring at lag ℓ independently; the contributions with $\ell > t - u$ are untouched by the conditioning and pool by Poisson superposition into $\text{Poisson}(R^* \Lambda(t))$. For SSE the same contributions are $\text{NB}(R^* w_\ell, k w_\ell)$ and pool by negative binomial closure into $\text{NB}(R^* \Lambda(t), k \Lambda(t))$, whose zero-probability is as stated. For SSI, given the latents they are again Poisson, with rates $R^* w_\ell Y_u$, pooling into $\text{Poisson}(R^* \Lambda_Y(t))$. For DLO, the future counts are negative binomial day by day with a dispersion that does not depend on the force of infection, so the zero-probabilities multiply over days. $\square$

(19) is included because it is the reference the other three are read against (Section 4.4), not because Cori is compared: it is the Poisson limit of (20) and (21) as $k \rightarrow \infty$, and $\Lambda_Y(t) \rightarrow \Lambda(t)$.

The DLO product is finite in practice, since $\mu_j = 0$ once j exceeds t plus the support of w . Under every infection-anchored model $\text{RAT} = \text{RAC}$, because an infection and the case it produces fall on the same day by construction.

4.3 Onset-anchored models

Here the observed onsets do not determine the state needed to project forward: at day t there is a latent pipeline of individuals already infected but not yet symptomatic. Both quantities nevertheless have closed forms conditional on a posterior draw, and it is these that are evaluated — no forward simulation is required. Fix a draw and write R_u for the historical reproduction number on day u and R^* for the reset value. Define three quantities.

Retained transmission weight. The driving series still able to transmit after t ,

$$W(t) = \sum_{u \leq t} \mathcal{D}_u \bar{f}_{t-u}^{\text{tost}}, \quad \mathcal{D}_u = \begin{cases} D_u & (\text{Cori-SO, SSE-SO}), \\ Y_u & (\text{SSI-SO}), \end{cases} \quad (23)$$

the onset-anchored analogue of $\Lambda(t)$, with the TOST survival function of (1) in place of $1 - F$.

Incubation pipeline mean. The expected number infected on or before day t whose onset falls after it,

$$M(t) = \sum_{u \leq t} E_u \bar{f}_{t-u}^{\text{inc}}, \quad (24)$$

with E_u the model's conditional mean number of infections on day u — $R_u \lambda_u$ for Cori-SO, $R_u \tilde{\lambda}_u$ for SSE-SO and $R_u \lambda_u^Y$ for SSI-SO, as in (12)–(16) — evaluated at the *historical* R_u : the pipeline was filled before the reset.

Per-case exponent. Minus the log-probability that one future case, transmitting freely under R^* over its whole TOST support, has no offspring at all,

$$c = \begin{cases} R^* & (\text{Cori-SO}), \\ k \log(1 + R^*/k) & (\text{SSE-SO, SSI-SO}). \end{cases} \quad (25)$$

Proposition 3 (Conditional zero-probabilities, onset-anchored models). *Let*

$$H(t) = \begin{cases} R^* W(t) & (\text{Cori-SO, SSI-SO}), \\ c W(t) & (\text{SSE-SO}). \end{cases} \quad (26)$$

Then, conditional on one draw of the parameters and the retained latent state,

$$\log \Pr(\text{no further case after } t) = -H(t) - M(t), \quad (27)$$

$$\log \Pr(\text{no further transmission after } t) = -H(t) - M(t)(1 - e^{-c}). \quad (28)$$

Proof. Two independent sources of future events must be accounted for: the onset cohorts retained through day t , which have transmission opportunity left, and the pipeline of individuals already infected but not yet symptomatic.

Retained cohorts. Under Cori-SO and SSI-SO the infections generated by cohort u at TOST lag s are independent $\text{Poisson}(R^* f_s^{\text{tost}} \mathcal{D}_u)$; those with $s > t - u$ are untouched by the conditioning, and summing over s and $u \leq t$ gives $\text{Poisson}(R^* W(t))$, with zero-probability $e^{-R^* W(t)}$. Under SSE-SO the same contributions are $\text{NB}(R^* f_s^{\text{tost}} D_u, k f_s^{\text{tost}} D_u)$ with common success probability $k/(k + R^*)$, so they pool into $\text{NB}(R^* W(t), kW(t))$, whose zero-probability is $(1 + R^*/k)^{-kW(t)} = e^{-cW(t)}$. Either way the probability is $e^{-H(t)}$.

Pipeline. Poisson splitting of the day- u infections by the incubation delay (the argument of Appendix A) makes the number infected on or before t but symptomatic after t Poisson with mean $M(t)$, independently of the retained cohorts' remaining transmission.

A future *case* requires either a pipeline member to become symptomatic or a retained cohort to generate a new infection — and every new infection produces an onset strictly later, since f^{inc} has no mass at lag 0. So no further case has probability $e^{-M(t)} e^{-H(t)}$, giving (27).

A future *transmission event* permits pipeline members to become symptomatic provided each transmits zero times, which each does independently with probability e^{-c} by (25). If the pipeline size is $N \sim \text{Poisson}(M(t))$ then $\mathbb{E}[(e^{-c})^N] = \exp(-M(t)(1 - e^{-c}))$ by the Poisson generating function, and multiplying by $e^{-H(t)}$ gives (28). $\square$

Treating the three models together is convenient for the derivation, but the quantity each one actually evaluates is a single expression in the observed onsets and the retained latent path. Substituting (23)–(26) into Proposition 3 gives, with λ_u , λ_u^Y and $\tilde{\lambda}_u$ as in Section 3.3:

$$\text{Cori-SO : } \Pr(\text{no case}) = \exp\left(-R^* \sum_{u \leq t} D_u \bar{f}_{t-u}^{\text{tost}} - \sum_{u \leq t} R_u \lambda_u \bar{f}_{t-u}^{\text{inc}}\right), \quad (29)$$

$$\Pr(\text{no transmission}) = \exp\left(-R^* \sum_{u \leq t} D_u \bar{f}_{t-u}^{\text{tost}} - (1 - e^{-R^*}) \sum_{u \leq t} R_u \lambda_u \bar{f}_{t-u}^{\text{inc}}\right); \quad (30)$$

$$\text{SSE-SO : } \Pr(\text{no case}) = \left(1 + \frac{R^*}{k}\right)^{-k \sum_{u \leq t} D_u \bar{f}_{t-u}^{\text{tost}}} \exp\left(-\sum_{u \leq t} R_u \tilde{\lambda}_u \bar{f}_{t-u}^{\text{inc}}\right), \quad (31)$$

$$\Pr(\text{no transmission}) = \left(1 + \frac{R^*}{k}\right)^{-k \sum_{u \leq t} D_u \bar{f}_{t-u}^{\text{tost}}} \exp\left(-\left[1 - \left(1 + \frac{R^*}{k}\right)^{-k}\right] \sum_{u \leq t} R_u \tilde{\lambda}_u \bar{f}_{t-u}^{\text{inc}}\right); \quad (32)$$

$$\text{SSI-SO : } \Pr(\text{no case}) = \exp\left(-R^* \sum_{u \leq t} Y_u \bar{f}_{t-u}^{\text{tost}} - \sum_{u \leq t} R_u \lambda_u^Y \bar{f}_{t-u}^{\text{inc}}\right), \quad (33)$$

$$\Pr(\text{no transmission}) = \exp\left(-R^* \sum_{u \leq t} Y_u \bar{f}_{t-u}^{\text{tost}} - \left[1 - \left(1 + \frac{R^*}{k}\right)^{-k}\right] \sum_{u \leq t} R_u \lambda_u^Y \bar{f}_{t-u}^{\text{inc}}\right). \quad (34)$$

In each pair the first sum is the transmission still owed by onset cohorts observed through day t , and the second is the incubation pipeline. The two differ only in the factor multiplying the pipeline term: a further *case* needs the pipeline to be empty, whereas a further *transmission event* needs only that no pipeline member goes on to infect anyone. Deleting the pipeline term from (31) and (33) leaves exactly the SSE and SSI expressions of Proposition 2, with the onset cohorts D_u in place of the infection cohorts I_u and the TOST survival function in place of $1 - F$ — which is the precise sense in which the infection-anchored models assume the pipeline away.

Remark (Consequences worth stating). Three special cases follow immediately and are used as verification targets (Section 5.6).

- (i) Since $1 - e^{-c} \leq 1$, (28) is never below (27), so $\text{RAC}(t) \geq \text{RAT}(t)$ always, with equality exactly when $M(t) = 0$. The gap is the pipeline, and it is the term the infection-anchored models set to zero by assumption.
- (ii) Setting $R^* = 0$ gives $c = 0$ and $H(t) = 0$, so $\text{RAT}(t) = 0$ exactly and $\text{RAC}(t) = 1 - e^{-M(t)}$: with no future transmission permitted, the only remaining risk is that the retained pipeline is non-empty.
- (iii) As $k \rightarrow \infty$, $c = k \log(1 + R^*/k) \rightarrow R^*$ and SSE-SO's $H(t)$ tends to $R^*W(t)$, so both overdispersed models collapse onto Cori-SO.

Remark (Where the reset asymmetry enters). $H(t)$ uses the reset R^* , because it describes transmission that has yet to happen; $M(t)$ uses the historical R_u , because the pipeline it counts was generated before the reset. Under the naive models there is no pipeline, so the whole retained state is projected at R^* . This is the formal counterpart of the timing argument of Section 6.3.

4.4 What the three mechanisms do differently

Proposition 2 makes the structural comparison explicit. SSE and SSI depend on the observed history *only through the scalar* $\Lambda(t)$ or $\Lambda_Y(t)$: the total remaining offspring pools into a single negative binomial or, given the latents, a single Poisson. DLO does not. Its risk depends on the whole temporal *profile* $(\mu_j)_{j>t}$ of the future force of infection, because a fresh dispersion that does

not scale with that force is applied on each future day. Writing $\varphi(\mu) = -k \log(1 + R\mu/k)$, which is convex with $\varphi(0) = 0$ and hence superadditive,

$$-R\Lambda(t) \leq \log \Pr(\text{no further case})_{\text{DLO}} = \sum_{j>t} \varphi(\mu_j) \leq \varphi(\Lambda(t)). \quad (35)$$

Spreading the same total $\Lambda(t)$ across days *dilutes* the overdispersion, so a value of k calibrated as an individual-level offspring dispersion does far less work in DLO than in SSE or SSI. The lower bound is exactly the Poisson limit (19), so DLO’s risk is squeezed towards Cori’s precisely when the future force of infection is thin and spread out.

Remark. The bounds (35) hold always; the *direction* of the comparison with SSE does not. It is a property of a thin, spread-out profile rather than a theorem, and concentrating $\Lambda(t)$ into a day or two reverses it.

4.5 Latent states at the conditioning day

For the latent-variable models, the risk at day t in principle wants latents conditioned on the history through t — a *filtering* distribution — whereas a single full-data fit delivers the *smoothed* distribution given the whole record.

We use the full-data (smoothed) posterior for every t . One fit per model then serves every day of every curve, which keeps each analysis to a handful of posterior simulations rather than several hundred. This is consistent with the analysis being retrospective by design: the parameters are estimated from all the data, and the latents are treated the same way. The consequence should be stated plainly — the projection at day t uses latents informed by data after day t , so $\text{RAC}(t)$ is not literally conditioned on the history through t for SSI, SSE-SO and SSI-SO. DLO and SSE are unaffected, having no latents.

The *direction* of the effect is a hypothesis rather than an assertion. Smoothing sharpens the latents but does not push them uniformly one way: observations of zero after day t are evidence that recent cohorts had low infectivity and pull the retained state down, while positive observations after day t are evidence that earlier cohorts had high infectivity and pull it up. After the final onset every subsequent observation is a zero, so the expectation is a predominantly downward effect over the decision-relevant late period and an ambiguous one earlier.

We measured this rather than assuming it, using the filtering distributions produced as a by-product of a particle filter’s forward pass. Both signs appear, and where predicted. For SSI at a fixed parameter vector, smoothed minus filtering risk is +0.64 around day 2, where the filter has not yet seen the cases that reveal a high early infectivity, and is negative after the last observed case, with mean -0.031 and worst -0.153 around day 70. For the onset-anchored models the same comparison, taken in the opposite direction, gives filtering minus smoothed risk of about -0.63 to -0.66 at its largest early and a mean of about $+0.013$ after the final onset. The late-period effect is small enough not to move the threshold crossings by more than about a day, but it is a signed, measured quantity rather than a claim.

4.6 Recovering integrated-out latent states

The latent parameterisation of Section 5.3 integrates out, in closed form, every latent that the data constrain only through a multiplicative $\exp(-\sum_j \mu_j)$ factor. On this series those are exactly the days from the last observed onset onwards — precisely the days a late conditioning day needs for its incubation pipeline. They are therefore drawn back from their exact conditional $\text{Gamma}(k \text{ scale}_u, k + c_u)$ and spliced onto the sampled block before any risk is computed, giving a complete latent path by day. This is exact, since the removed latents are conditionally independent of the sampled ones and of each other, and it costs one Gamma draw per removed latent per posterior draw. Omitting it would truncate the retained state at day 58 and understate the risk over exactly the period in which the decision is made.

5 Inference

5.1 Priors

Priors are proper and identical across the models within an analysis. This is not a stylistic choice: the posterior model probabilities are Bayes factors, and diffuse or improper priors would make them arbitrary.

- R_{pre} and R_{post} : independent log-normals with median 1.0 and 95% interval (0.2, 5.0).
- k , where estimated: a deliberately fairly informative log-normal with median 0.18 and 95% interval (0.09, 0.36), identical in every model.
- Prior model probabilities: uniform over the models in each analysis, i.e. 1/3 in Analyses 1–2 and 1/4 in Analyses 3–4.

Remark (Identical priors are necessary but not sufficient for a fair comparison). Propriety makes the Bayes factors well defined; it does not make them *neutral* when the same symbol means different things in different models. k is exactly such a symbol: an individual-level offspring dispersion in SSE and SSI, a day-level incidence dispersion in DLO (Section 4.4). Placing the same prior on both is a deliberate choice. Analysis 1 in particular compares models one of which is *deliberately miscalibrated*, in order to demonstrate the error under study, so its posterior model probabilities are evidence about those specified models rather than clean evidence about mechanism. Prior-predictive calibration is out of scope here, so this caveat carries the weight instead.

5.2 Posterior sampling

All fits use Hamiltonian Monte Carlo (the No-U-Turn sampler, as implemented in PyMC), with 4 chains of 2000 draws after 2000 tuning iterations in every analysis, at a target acceptance probability of 0.90 in Analyses 1–3 and 0.95 in Analysis 4. One fit per (model, analysis) pair jointly estimates R_{pre} , R_{post} , any latents, and — in Analyses 2 and 4 — k .

SSE and SSI appear in both the infection-anchored and the onset-anchored analyses. They are refitted independently in each, so their summaries can differ between figures by Monte-Carlo error; no number is carried across.

Convergence. Across all models of all four analyses, the worst $\hat{R}$ for the reported scalar parameters is 1.00, the smallest bulk effective sample size is 3575, and the total number of divergent transitions is 0. Per analysis the minimum bulk effective sample sizes are 4855, 4074, 4190 and 3575, with no divergences in any of them.

5.3 The latent block

Sampling the latent Gammas directly does not work. The SSE-SO shapes on this series span several orders of magnitude, down to around 10^{-6} , and a Gamma coordinate with a shape that small has a typical value outside the range of double precision, so no amount of tuning or reinitialisation will make it behave. Two devices are used, and they compose.

- **Exact marginalisation.** A latent that reaches no observation day carrying a case enters the likelihood only through $\exp(-\sum_j \mu_j)$, which is linear in the latents and so factorises. Its Gamma prior is conjugate to that factor, giving $\int \text{Gamma}(y; k \text{ scale}, k) e^{-cy} dy = (1 + c/k)^{-k \text{ scale}}$ and a conditional posterior $\text{Gamma}(k \text{ scale}, k + c)$. On this series that removes 52 of SSE-SO’s 110 latents, leaving 58 sampled — exactly the pathological tail, since the last case is on day 58 — and 1 of SSI’s and SSI-SO’s 31. It is exact, and Section 4.6 describes how the removed latents are recovered where the risk calculation needs them.

- **Inverse-CDF reparameterisation.** The surviving latents are sampled as $\text{Uniform}(0, 1)$ variates pushed through the Gamma quantile function, giving a bounded, flat-prior, constant-geometry parameter space. The quantile function is differentiable in the automatic-differentiation framework used, so this runs under the same gradient-based sampler as everything else.

Remark. The choice rests on a benchmark of 74 posterior simulations, whose main negative finding constrains what else might be tried: this is a geometry problem, not a warm-up problem. Rescaling the latents to unit mean — a translation on the log scale, which leaves the curvature untouched — leaves the SSE-SO fit with essentially every draw divergent, exactly as the untouched model does, and additional tuning makes $\hat{R}$ worse rather than better. The inverse-CDF reparameterisation takes the same model from 4000 divergences to none.

5.4 Model evidence

The marginal likelihood $p(D_{1:110} \mid \text{model})$ is computed with the parameters *and* the latents integrated out, from each model’s own joint density, so that the integral estimated is the one that was sampled. The integral is taken in unconstrained coordinates with the Jacobian included. Bridge sampling is the estimator; prior Monte Carlo and importance sampling are retained as agreement checks, and deterministic quadrature as the exact answer where the parameter space is small enough. Posterior model probabilities follow from the evidences under the uniform model prior of Section 5.1.

5.5 Comparing dispersion posteriors

Where k is estimated, the models are given the same prior on it, so wherever their posteriors end up apart is a measurement rather than a demonstration. Three complementary quantities are reported for every pair, because none says enough alone: the *median ratio* (location, but not width); the *overlap* $\int \min(p_a, p_b)$, which is scale-free, symmetric and finite where a Kullback–Leibler divergence would not be; and $\Pr(k_a > k_b)$ for independent draws. All are computed on the log scale, as the density panels are drawn; the overlap is invariant to that choice, since both densities acquire the same Jacobian. Every pair is compared rather than a designated reference model.

5.6 Verification

The risk calculation is verified against three independent routes, and the models against their own limits.

Limits and equivalences. Setting $f_1^{\text{inc}} = 1$ (a deterministic one-day incubation) and shifting indices recovers the infection-anchored models from the onset-anchored ones; $k \rightarrow \infty$ collapses SSE and SSI onto Cori, and SSE-SO and SSI-SO onto Cori-SO; forward-simulating the natural and inference-convenient forms of the onset-anchored models gives matching distributions of the observed onsets (Appendix A); and the inference-convenient likelihood matches brute-force marginalisation over the latent infection counts on short histories. The closed forms of Proposition 3 are checked directly against explicit reset simulation, and, in the degenerate case $R^* = 0$, against the statement that RAT is exactly zero and RAC exactly the probability that the retained pipeline is non-empty.

Particle filtering and particle MCMC. Two checks answer different questions. At fixed $(R_{\text{pre}}, R_{\text{post}}, k)$, with conditioning matched to the pipeline, a particle filter provides an equality check on the risk arithmetic; agreement is obtained to 3.2×10^{-3} for SSI and 0.021 for the onset-anchored models. Over the parameters, particle marginal Metropolis–Hastings gives an independent route to the posterior sharing no machinery with the main fits — the strongest

available guard against an error in the latent parameterisation, which simulation-based calibration could not catch, since it validates an implementation against itself. Posterior intervals from the two routes overlap for all six synthetic parameter comparisons, and the largest measured variance of the estimated log-likelihood is 0.081 at 250 particles, comfortably inside the range particle MCMC tolerates. A particle-MCMC risk curve should *not* match the main one for the latent models: the filter conditions on filtering latents where the main calculation uses smoothed ones, and that gap is the Section 4.5 measurement rather than a defect.

Evidence estimation. Bridge sampling reproduces deterministic quadrature for DLO and SSE on this series to 1.1×10^{-3} and 2.9×10^{-3} nats, with the quadrature box’s boundary density 41–44 nats below its peak; the three estimators agree on SSI, where no exact answer exists, all within their own error bars; and the evidence is invariant to the latent parameterisation, which is what shows the marginalisation carries the whole removed factor rather than just its shape.

External replication. In the Poisson limit, the risk calculation reproduces the published Équateur risk curve of Thompson et al. (2024) exactly under their conventions, which differ from ours by exactly one day. Their published reproduction-number posterior for this outbreak is recovered, and the two risk curves agree to 2.2×10^{-4} over 200 000 draws.

6 Results

Four analyses are reported. Analyses 1 and 2 compare the three infection-anchored models, with k held at the literature value 0.18 and then estimated; Analyses 3 and 4 add the onset-anchored models to SSE and SSI, again fixed and then estimated. Throughout, “the crossing” means the first day on which a curve falls below a threshold *and stays there*. Values quoted at day 90 (4 July 2018) are read from the fitted curves partway down their descent, where they are furthest apart.

6.1 Mechanism, at a transplanted dispersion parameter

- **The same k buys DLO almost no overdispersion.** At day 90 the risk is 0.413 under DLO against 0.093 under SSE and 0.123 under SSI — a factor of more than four between models handed an identical dispersion parameter. This is the dilution bound (35) in practice: the future force of infection after the last case is thin and spread over the whole serial interval, which is exactly the regime in which a day-level k does almost nothing and DLO is pressed against its Poisson lower bound.
- **The consequence for a declaration is weeks, not days.** DLO’s risk first settles below 0.05 on day 107 (21 July 2018), against day 95 for SSE and day 97 for SSI. At the 0.01 threshold DLO **never** settles within the analysis window, while SSE does so on day 106 and SSI on day 108. On the day the ERT actually withdrew, DLO still places 0.032 on a further case, against 0.005 and 0.007.
- **SSI lies on the same side of SSE as DLO does.** After the final case SSI’s risk is *above* SSE’s (0.123 against 0.093 at day 90), which reads backwards until one asks what each model’s heterogeneity is attached to. SSI’s is a property of *people who have been observed*: 54 known individuals have now gone weeks without infecting anyone, which is direct evidence they were not the superspreading kind, and $\Lambda_Y(t)$ is revised down accordingly. SSE’s is a property of *events that have not happened yet*, about which a run of zeros says nothing, so SSE gets no such revision — its entire retained state is $\Lambda(t)$. At matched expected further cases, SSE’s residual risk is a rare burst and SSI’s a more likely singleton, and the question “any further case?” sees the second as the larger risk. So the axis is not naive-versus-mechanistic: it is how many units the same $\Lambda(t)$ is divided over and whether each unit’s dispersion survives to time t . DLO divides over future days, SSI over past individuals, SSE not at all.

Équateur 2018: naive (onset-as-infection) models, k fixed at 0.18

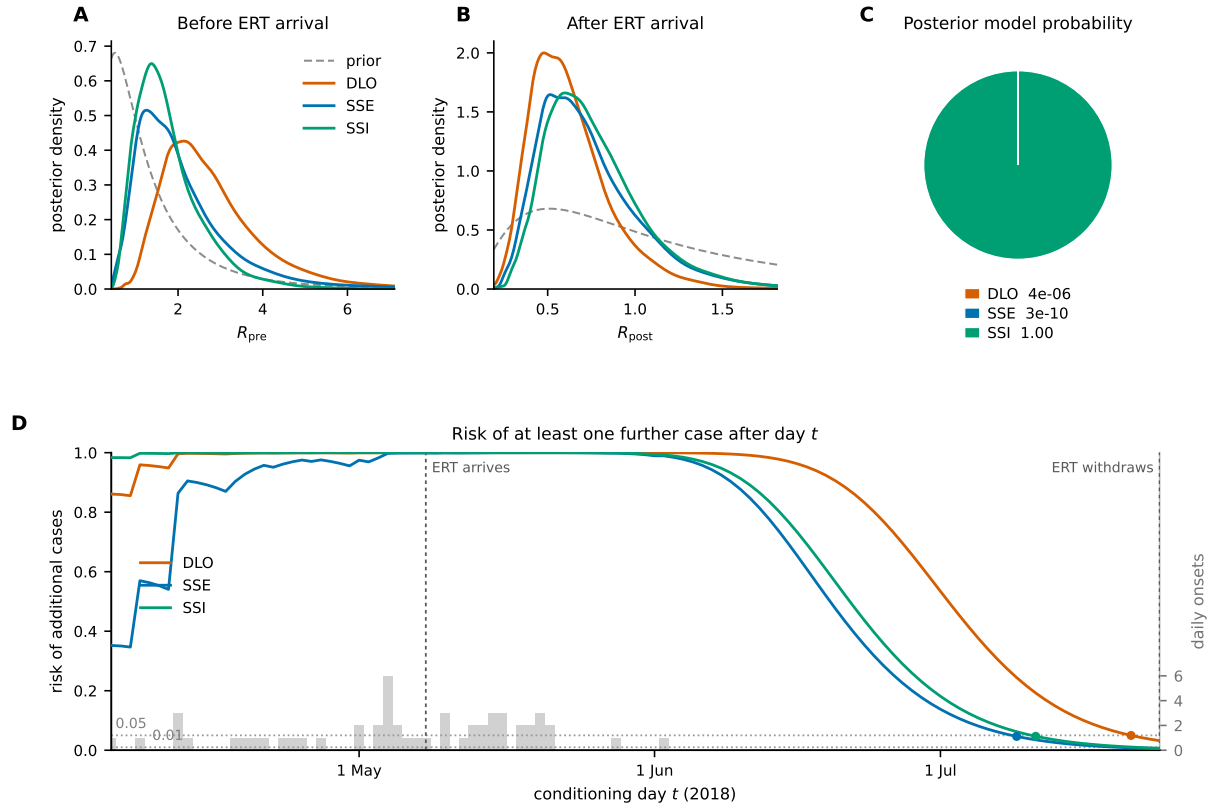


Figure 1: The mechanism of overdispersion changes the end-of-outbreak risk, at a dispersion parameter that is only meaningful for two of the three models. DLO, SSE and SSI fitted to the Équateur onset series with k fixed at 0.18. **A**, **B**: posterior densities for R_{pre} and R_{post} , over the shared prior. **C**: posterior model probabilities under a uniform prior over the 3 models; one of the three is deliberately miscalibrated, so this panel is evidence about the models *as specified*, priors included. **D**: risk of additional cases against conditioning day, with daily onsets behind it, the ERT arrival and withdrawal marked, and the 0.05 and 0.01 decision thresholds drawn. The marker on each curve is its 0.05 crossing.

- **The ordering is a property of the question as much as of the models.** Asking instead whether transmission would re-establish reverses SSE and SSI on these same data, because a cluster seeds a new chain far more readily than a lone case. Both orderings are correct answers to different questions, and the end-of-outbreak criterion in use asks the first.
- **The model probabilities strongly favour SSI** — 1.00 against 4×10^{-6} for DLO and 3×10^{-10} for SSE, from log evidences of -84.8 , -97.1 and -106.8 nats. At a fixed k this partly reports that 0.18 is a better value for SSI than for the other two, which is the transplantation error restated rather than independent evidence about mechanism. Analysis 2 separates the two readings.
- **R_{post} is comparable across the three models** — 0.59 (0.30–1.20), 0.68 (0.34–1.48) and 0.72 (0.36–1.48) (posterior median and 95% credible interval) — so the divergence in the risk curves does not come from a different inferred effect of the intervention. The corresponding R_{pre} estimates are 2.55 (1.20–6.05), 1.81 (0.72–5.35) and 1.61 (0.73–3.87).
- **Monte-Carlo error does not account for the separation.** Over the case-free tail in which the crossings occur, the largest Monte-Carlo standard error on any of the three curves is 0.0015, two orders of magnitude below the differences above.

6.2 Mechanism, with the dispersion estimated

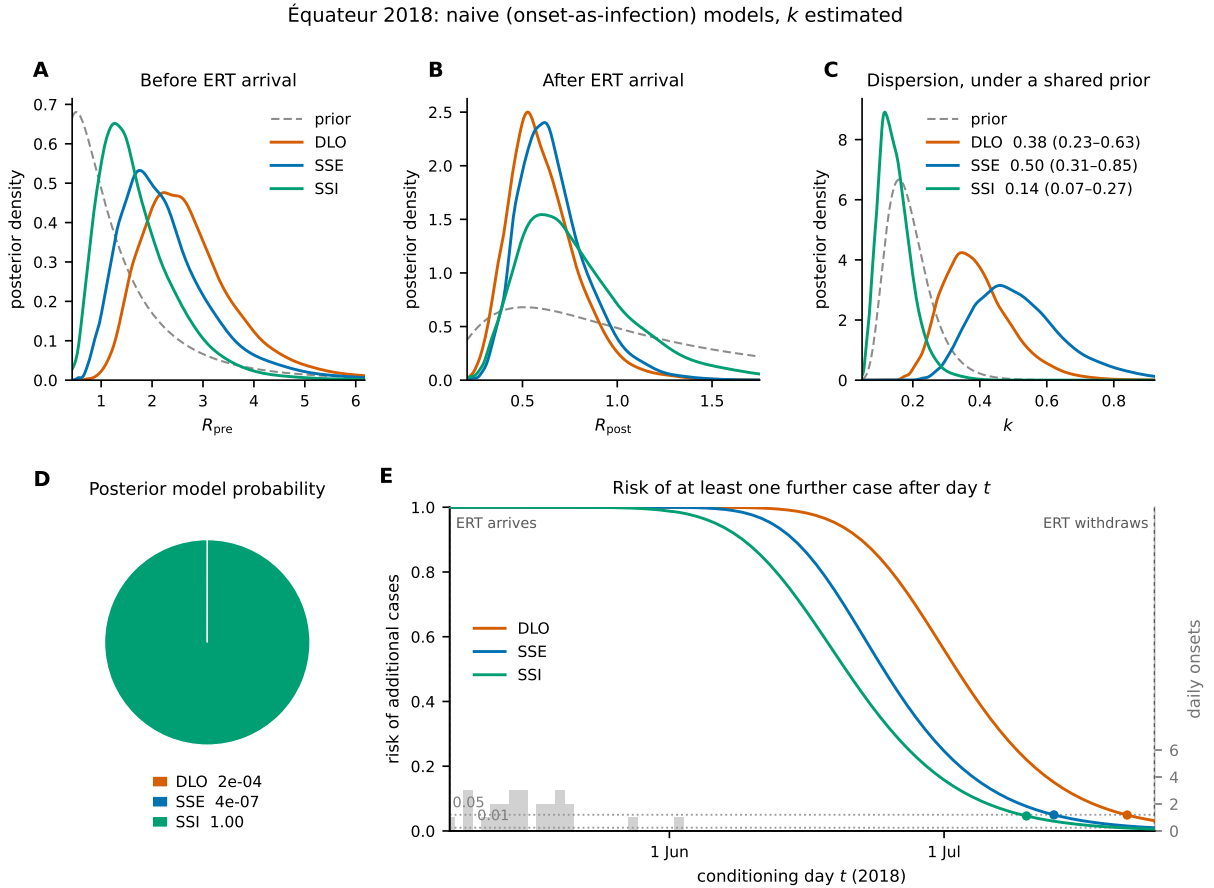


Figure 2: Estimating k under a shared prior shows that the models want different values of it — and that the risk gap survives anyway. As Fig. 1, with k estimated. **C:** the three k posteriors over the prior they share, with each model's posterior median and 95% credible interval in the legend. **D:** posterior model probabilities. **E:** risk of additional cases, with the view starting at the ERT's arrival because all three curves sit flat at 1 until well past the last case.

- **The models disagree about k , but not along the axis one might expect.** The posterior medians (95% credible intervals) are 0.378 (0.230–0.629) for DLO, 0.502 (0.307–0.853) for SSE and 0.141 (0.072–0.271) for SSI. DLO against SSI: overlap 0.094, median ratio 2.69, $\Pr(k_{\text{DLO}} > k_{\text{SSI}}) = 0.990$. SSE against SSI: overlap 0.032, ratio 3.57, probability 0.999. But DLO against SSE: overlap 0.589 and $\Pr(k_{\text{DLO}} > k_{\text{SSE}}) = 0.219$ — the two are barely distinguishable. **It is SSI that is the outlier, not DLO.**
- **The split is day-level against individual-level, which is where onset-as-infection bites hardest.** DLO and SSE both attach their excess variance to a *day*: DLO to the day’s aggregate incidence, SSE to the day’s pooled transmission through a freshly drawn λ_t . SSI attaches it to the *individual*. The series being fitted is onsets, and an incubation period of mean 11.4 d and standard deviation 8.1 d scatters each day’s infections forward over a wide kernel — so a model reading onsets as infections sees day-to-day variation that the convolution has already largely averaged out, and answers with a larger k (less overdispersion) than the infection process carried. SSI is close to invariant to this, because the convolution *regroups* individuals across days while the aggregate infectivity of n independent individuals is $\text{Gamma}(kn, k)$ whichever n they are: regrouping moves the cohorts, not the variance. This is a proposed mechanism for the measurement, not a second measurement — these three models are all infection-anchored, so this analysis cannot separate it from the purely structural difference of Section 4.4. Section 6.3 is where it is tested.
- **Only SSI’s posterior is compatible with the literature value.** SSI’s 95% credible interval, (0.072, 0.271), contains 0.18; DLO’s (0.230, 0.629) and SSE’s (0.307, 0.853) do not, and both medians sit above the prior’s own 97.5th percentile of 0.36, despite that prior being deliberately informative.
- **The evidence says the same thing more sharply.** Letting k move is worth +7.41 nats to SSE and +3.90 to DLO, but only +0.26 to SSI. Analysis 1 was charging DLO and SSE for a value that was never theirs, and 0.18 really is SSI’s number on this series.
- **The risk gap survives, which is the point of the pairing.** Each model now sits at its own best k , and DLO still crosses 0.05 on day 107, against day 99 for SSE and day 96 for SSI, and still **never** reaches 0.01 inside the window, where SSE does on day 110 and SSI on day 107. Misapplying a literature k creates the gap; the day-level mechanism sustains it.
- **The posterior model probabilities keep their direction** — SSI 1.00, DLO 2×10^{-4} , SSE 4×10^{-7} — and now carry a cleaner reading, since each model has been allowed its own dispersion.

6.3 Onset anchoring

- **Onset anchoring advances both crossings by 8–12 days.** With k estimated (Fig. 4), SSE-SO crosses 0.05 on day 87 against SSE’s day 99, an advance of 12 days, and 0.01 on day 98 against day 110, an advance of 12 days. SSI-SO advances SSI’s crossings by 8 and 8 days respectively (day 88 against day 96, and day 99 against day 107). At a fixed k (Fig. 3) the same comparisons give 7 and 7 days for SSE-SO against SSE, and 8 and 9 days for SSI-SO against SSI. The displacement is on the scale of the mean incubation period, 11.4 d.
- **Why the direction is *earlier*, and why it is not a shorter generation time.** All four models share the same onset-to-onset serial interval; what changes is which side of the conditioning day a transmission falls on once that interval is cut into transmission and incubation pieces. Proposition 3 makes this explicit: the onset-anchored models place infection at transmission time, so infections already generated under the historical R_{post} enter $M(t)$ and stay in the incubation pipeline when control is reset, while the remaining TOST opportunity $W(t)$ of older onset cohorts is already largely spent. The infection-anchored models have no pipeline, move the infection decision to the later symptom-onset date, and apply the reset R^* to it, which makes the residual state look younger and drain later.

Équateur 2018: onset anchoring compared, k fixed at 0.18

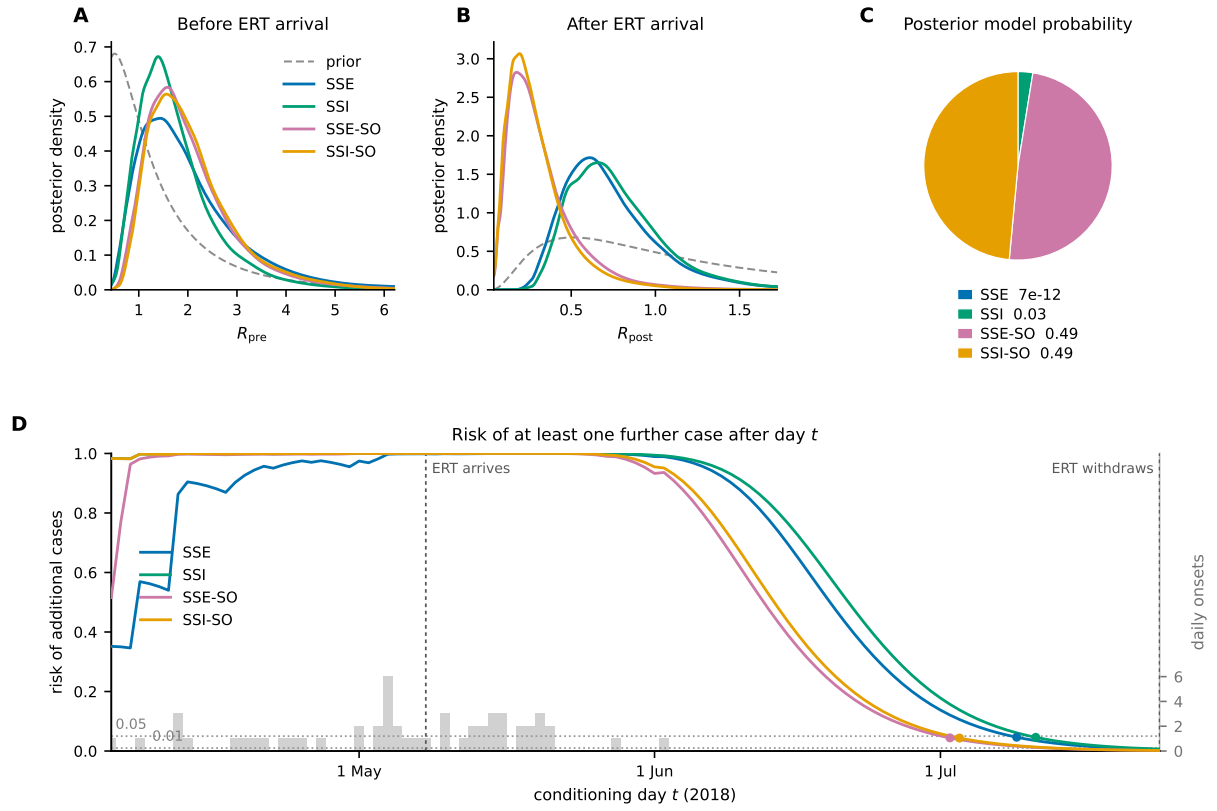


Figure 3: Onset anchoring at a fixed dispersion parameter. SSE, SSI, SSE-SO and SSI-SO with k held at 0.18; panels as in Fig. 1, with a uniform prior over 4 models in C. All four models are driven by the same onset-to-onset serial interval, so the comparison isolates anchoring.

Équateur 2018: onset anchoring compared, k estimated

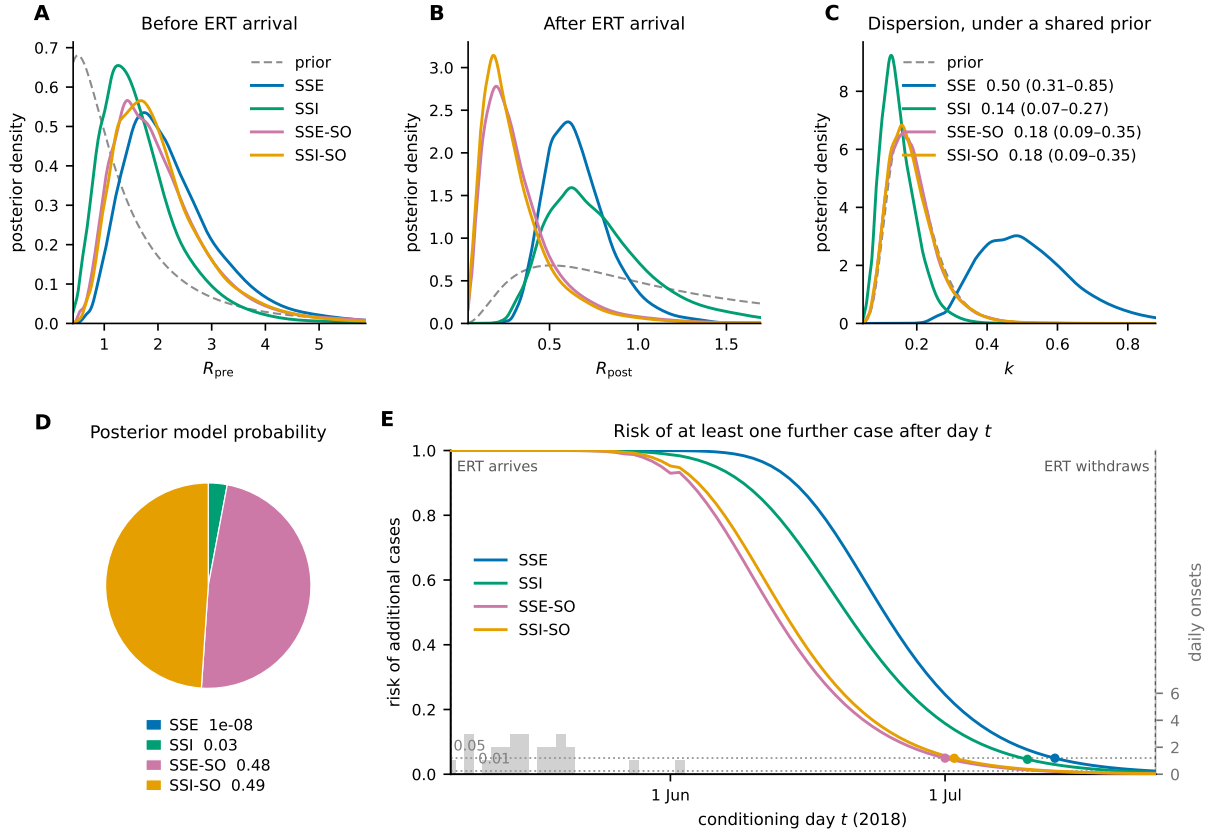


Figure 4: Onset anchoring with the dispersion estimated. Panels as in Fig. 2. Panel C is where the attenuation hypothesis of Section 6.2 is settled: SSE-SO’s k posterior should sit below naive SSE’s if incubation convolution is what inflates day-level dispersion estimates, while SSI-SO’s should barely move from SSI’s.

- **The same indexing error inflates the naive R_{post} roughly threefold.** With k estimated the posterior medians (95% credible intervals) are 0.63 (0.37–1.11) for SSE and 0.73 (0.35–1.60) for SSI, against 0.28 (0.08–0.95) for SSE-SO and 0.26 (0.08–0.89) for SSI-SO. The naive convention applies the day-33 switch to *onsets*, so post-arrival onsets that arose from pre-arrival infections are charged to the controlled period. R_{pre} is far more stable across the four — 2.09 (0.98–4.77), 1.54 (0.65–3.67), 1.86 (0.84–4.61) and 1.87 (0.87–4.41) — which is what one expects if the effect concerns *when* the switch lands rather than the epidemic’s early growth.
- **The attenuation hypothesis passes.** SSE-SO’s k posterior, 0.178 (0.088–0.351), sits below naive SSE’s 0.501 (0.306–0.847) and towards the literature value: median ratio 2.81, overlap 0.089, $\Pr(k_{\text{SSE}} > k_{\text{SSE-SO}}) = 0.992$. SSI moves far less — 0.140 (0.072–0.271) against SSI-SO’s 0.175 (0.088–0.353), overlap 0.750. Incubation convolution attenuates *day-level* dispersion when onsets are read as infections; individual-level dispersion is comparatively stable under regrouping, as predicted.
- **Once anchoring is explicit, the mechanism stops mattering for k .** SSE-SO and SSI-SO agree almost exactly — overlap 0.980, median ratio 1.02, $\Pr(k_{\text{SSE-SO}} > k_{\text{SSI-SO}}) = 0.505$ — where their naive counterparts differ by a factor of 3.58 at an overlap of 0.034. The event-versus-individual gap of Section 6.2 was to a large extent the conflation, not the mechanism.
- **The onset-anchored models are strongly preferred, and are indistinguishable from each other.** With k estimated the posterior model probabilities are 0.48 for SSE-SO and 0.49 for SSI-SO, against 0.03 for SSI and 1×10^{-8} for SSE; the log evidences are -81.8 , -81.8 , -84.6 and -99.4 nats. The fixed- k analysis gives the same ordering (0.49, 0.49, 0.03, 7×10^{-12}).
- **The onset-anchored models gain nothing from estimating k , which is a consistency check on the two preceding points.** Releasing k is worth $+0.01$ nats to SSE-SO and $+0.04$ to SSI-SO — within their own estimation error — against $+7.41$ for naive SSE. Their k posteriors already sit on 0.18, so fixing it there costs them nothing.
- **The direction of the 8–12-day advance should not be universalised.** It is measured under this reset convention, this switch convention, this outbreak history and this posterior. The incubation-scale explanation is the mechanism for *this* result, not a theorem for arbitrary time series or interventions.

6.4 Cases against transmission: the incubation pipeline

- **The pipeline is worth about a quarter of a unit of probability at its widest.** With k estimated, RAC exceeds RAT by at most 0.269 for SSE-SO, on day 64 (8 June 2018), and by at most 0.240 for SSI-SO, on day 65. The gap peaks shortly after the last observed onset (day 58), when the pipeline is largest relative to the transmission still to come, and closes as it drains.
- **It delays a case-based declaration by most of a week.** With k estimated, SSE-SO’s RAT settles below 0.05 on day 81 and below 0.01 on day 92, against day 87 and day 98 for RAC — later by 6 and 6 days. For SSI-SO the same comparison gives 5 and 6 days (RAT on days 83 and 93 against RAC on days 88 and 99).
- **The two quantities answer different questions, and the difference is decision-relevant.** RAT isolates what relaxing interventions does; RAC is what a surveillance system would actually see. A criterion written on cases keeps interventions in place while the pipeline drains, even though transmission risk has already settled — and an infection-anchored model, which identifies the two by assumption, cannot represent that distinction at all.
- **Fixing k changes very little here,** as the near-identical panels show: the widest gap is 0.269 for SSE-SO and 0.236 for SSI-SO, against 0.269 and 0.240 when it is estimated — a

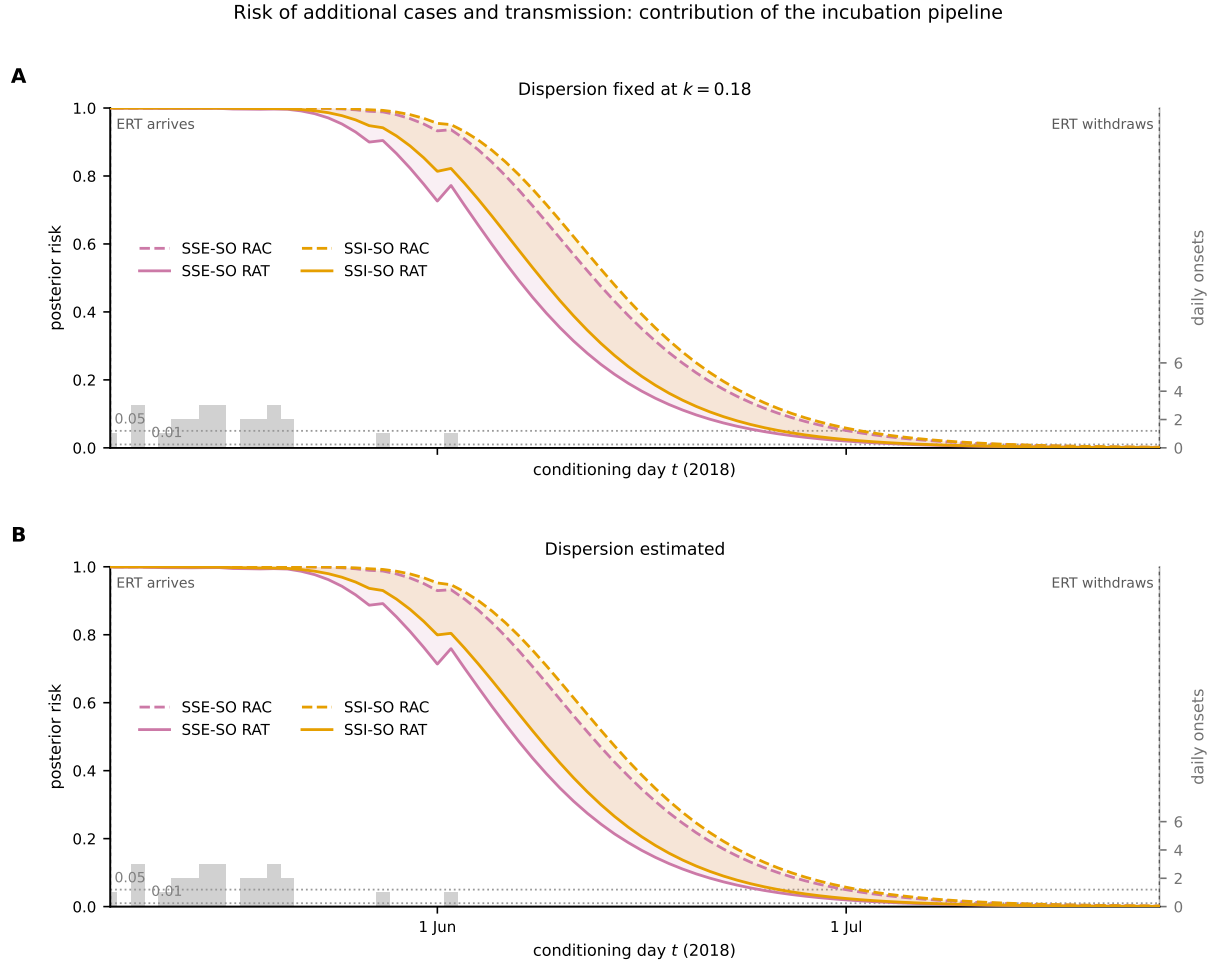


Figure 5: The gap between the risk of additional cases and the risk of additional transmission is the incubation pipeline, quantified. SSE-SO and SSI-SO with k fixed at 0.18 (A) and estimated (B). Dashed: RAC. Solid: RAT. The shaded region between them is the $M(t)e^{-c}$ term of Proposition 3 — individuals already infected on the conditioning day but not yet symptomatic. Under the three infection-anchored models these two curves coincide by assumption, which is the conflation being measured.

Table 1: Threshold crossings of the risk of additional cases across the four analyses: the first day on which the curve settles below the threshold, with the calendar date. The ERT withdrew on day 110 (24 July 2018); “never” means the curve had not settled below the threshold by then.

Analysis	Model	RAC < 0.05		RAC < 0.01	
		day	date	day	date
1: naive, $k = 0.18$	DLO	107	21 July 2018	never	never
	SSE	95	9 July 2018	106	20 July 2018
	SSI	97	11 July 2018	108	22 July 2018
2: naive, k estimated	DLO	107	21 July 2018	never	never
	SSE	99	13 July 2018	110	24 July 2018
	SSI	96	10 July 2018	107	21 July 2018
3: onset, $k = 0.18$	SSE	95	9 July 2018	106	20 July 2018
	SSI	97	11 July 2018	108	22 July 2018
	SSE-SO	88	2 July 2018	99	13 July 2018
	SSI-SO	89	3 July 2018	99	13 July 2018
4: onset, k estimated	SSE	99	13 July 2018	110	24 July 2018
	SSI	96	10 July 2018	107	21 July 2018
	SSE-SO	87	1 July 2018	98	12 July 2018
	SSI-SO	88	2 July 2018	99	13 July 2018

consequence of the onset-anchored k posteriors already sitting on the literature value.

7 Sensitivity analyses

None of the following has been run. They are listed in rough order of how much each would add, with what it would resolve.

1. **Outbreak-specific serial interval** (mean 19.46 d, standard deviation 6.08 d) — *infection-anchored models only*. Its much lighter tail materially lowers the risk after a long case-free period, and it is the estimate that should be preferred where outbreak-specific contact-tracing data exist. It is structurally inadmissible for the onset-anchored models (Section 2.2), and that incompatibility should be reported rather than worked around: it says something about how much structure the incubation/TOST decomposition imposes.
2. **Incubation/TOST decomposition**. Hold w fixed and re-run SSE-SO and SSI-SO under two or three other published Ebola incubation estimates, taking the matching TOST from the variance budget each time. This bounds how much of the onset-anchoring result rests on the particular split used here, which the serial interval alone does not identify.
3. **Filtering rather than smoothed latents**. Section 4.5 reports the signed comparison at a fixed parameter vector; repeating it integrated over the parameter posterior would turn the stated approximation into a quantified one for all three latent-variable models.
4. **Shifted R -switch for the infection-anchored models**. The onset-anchoring comparison confounds model structure with the indexing convention for the intervention. Re-running the naive models with the switch moved back by $\mathbb{E}[A] = 11.4$ days separates them: any residual difference from the onset-anchored models is then attributable to structure rather than to indexing.
5. **Non-empty initial incubation pipeline** for the onset-anchored models. Seed one or two already-infected individuals at day 0 with residual incubation drawn from f^{inc} , and measure

the effect on R_{pre} and on the risk curve. This is the sensitivity that matters if the index case is not epidemiologically identified (Section 2.1).

6. **Prior sensitivity**, in particular for k , whose prior is deliberately informative and whose role differs between models.

A The two forms of the onset-anchored models are equivalent

This appendix proves Proposition 1 and records the extra step (13) needs. Throughout, the natural form is as defined in Section 3.3: infections J_t are drawn from the model’s offspring distribution, and the J_u infections of day u are allocated to onset days by the incubation split (9), of which only the totals D_t are observed.

Proof of Proposition 1

By hypothesis $J_t \mid E_t \sim \text{Poisson}(E_t)$, with E_t measurable with respect to the history before day t .

Step 1 — Poisson thinning. Multinomial splitting of a Poisson count yields *independent* Poisson parts: conditional on E_u , $N_{u,a} \sim \text{Poisson}(E_u f_a^{\text{inc}})$ independently over $a \geq 1$, with $J_u = \sum_a N_{u,a}$ recovered as their sum. The latent J_u therefore carries no information beyond the parts and can be dropped. (This is also the step that makes the incubation pipeline of (24) Poisson.)

Step 2 — Poisson superposition. $D_t = \sum_{a \geq 1} N_{t-a,a}$ is a sum of independent Poissons, hence $D_t \mid (E_u)_{u < t} \sim \text{Poisson}(\sum_{a \geq 1} f_a^{\text{inc}} E_{t-a})$, which is (10).

Step 3 — the joint law, not just the marginal. Because f^{inc} has no mass at lag 0, E_{t-a} for $a \geq 1$ is a function of $D_{0:t-1}$ only, so the display above is precisely the conditional law $p(D_t \mid D_{0:t-1})$. The parts $\{N_{u,a} : u + a = t\}$ composing D_t are disjoint from those composing any other $D_{t'}$, so distinct days’ onsets are conditionally independent given the E s. Chaining $p(D_{0:T}) = \prod_t p(D_t \mid D_{0:t-1})$ gives equality of the full joint laws. $\square$

Why the hypothesis covers all three models

Cori-SO and SSI-SO satisfy it directly, with $E_t = R_t \lambda_t$ and $E_t = R_t \lambda_t^Y$ respectively; under SSI-SO the Gamma infectivities are explicit in both forms and the thinning and superposition arguments do not touch them.

SSE-SO needs one further step, because its offspring law (13) is negative binomial rather than Poisson. If $\tilde{\lambda}_t \mid \lambda_t \sim \text{Gamma}(k \lambda_t, k)$ and $J_t \mid \tilde{\lambda}_t \sim \text{Poisson}(R_t \tilde{\lambda}_t)$, then $R_t \tilde{\lambda}_t \sim \text{Gamma}(k \lambda_t, k/R_t)$ and, marginally, $J_t \sim \text{NB}(\text{mean} = R_t \lambda_t, \text{disp} = k \lambda_t)$ — which is exactly (13). So (14) merely *un-marginalises* the negative binomial into an explicit per-day Gamma random effect, after which J_t is Poisson conditional on $\tilde{\lambda}_t$ and the proposition applies with $E_t = R_t \tilde{\lambda}_t$. The $\tilde{\lambda}_t$ are independent across t given (λ_t) , and each λ_t is a function of $D_{\leq t}$, so the day- t conditional still depends only on $D_{0:t-1}$ once $\tilde{\lambda}_{< t}$ is integrated out. $\square$

B Code and reproducibility

The analysis is implemented in Python and orchestrated with Snakemake in three tiers — model fitting, derived quantities, and figures — so that a change at one tier does not re-run those above it. The committed repository contains the padded onset series, every posterior sample, every derived quantity, all figures and this document; `snakemake` reproduces all of them from the raw data.

No number in this document is typed into the prose. Each is a macro expanded from a generated file that a pipeline rule writes from the posterior samples and derived quantities, so a refit moves the numbers in the text as well as on the figures, and a key with no value is a compilation error rather than a blank. Quantities that are inputs rather than results — the serial

interval, the priors, the sampler settings — are generated from the configuration file in the same way.

The verification results quoted in Section 5.6 come from studies whose subject is the implementation rather than the outbreak, and whose full records are kept separately from the analysis outputs, in `validation/results/`: the sampler benchmark behind Section 5.3, the risk-calculation cross-checks and the smoothing comparison of Section 4.5, the marginal-likelihood checks of Section 5.4, and the onset-model particle-MCMC study.

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